# Vipul Yadav

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## Education

**Bipin Tripathi Kumaon Institute of Technology, Dwarahat** 2023 – 2027 (Expected)
*Bachelor of Technology in Computer Science & Engineering* *Uttarakhand, India*

## Experience

**Software Developer Intern** Dec 2025 – Sep 2026
*ColoredCow Consulting Pvt. Ltd.* *Remote*

- Owned critical workflows on a **blockchain-backed platform for secure access to sensitive documents** (Next.js, TypeScript, Tailwind CSS): traced each flow end to end, verified data stayed protected at every step, and drove gaps to fixes before release
- Built and fixed features including parts of the **secure sign-in flow** and user record/document workflows; integrated APIs with **smart contracts** and automated PDF generation
- Implemented core project-management, communication and **multi-language (i18n)** flows for an **AI-first enterprise project management system** for the international market, across several user personas
- Reduced bundle size by **~15%** by optimizing routing and middleware and removing dead code; built ColoredCow website pages using **AI-assisted development**
- Contributed across **24+ repositories** with **700+ GitHub contributions**, shipping client work on schedule

**Machine Learning Intern** Jun 2025 – Jul 2025
*Tamizan Skills India* *Remote*

- Built **4 end-to-end ML projects** (BERT emotion analysis, MNIST digit detection, resume matching, pose matching) using Python, PyTorch, Scikit-learn and OpenCV
- Ran the full ML workflow for each: data preprocessing, model training, hyperparameter tuning and evaluation across classification and computer vision tasks

## Projects

**MessLoop — AI Food-Waste Agent** | *TypeScript, AWS (Lambda, Step Functions, DynamoDB), Qwen3-4B* GitHub 2026

- Built for Environmental Hacks 2026: an event-driven agent (DynamoDB Streams → Lambda → Step Functions) that flags dishes wasted in 3 of their last 5 meals, has a self-hosted LLM explain the evidence, and **pauses for manager approval** before any change is applied
- Checked every AI output against code-computed numbers with a rule-based fallback, so a wrong answer never reaches the user; unit-tested against the spec's reference results (simulated data)
- Added Hindi voice logging (Amazon Polly + Transcribe) and SNS/EventBridge alerts; deployed with CloudFormation

**Nishaan — Decentralized Evidence Management System** | *Next.js, Solidity, Hardhat, IPFS, Ethers.js* Live 2026

- Built a Web3 evidence system for law enforcement with a verifiable chain of custody on Ethereum; **won 1st place at TechFest 2026**
- Encrypted files client-side (AES-256-GCM, per-reader sealed keys) before pinning to IPFS and anchored CID hashes in a Solidity contract with an on-chain audit log; the pinning service and admin cannot read evidence
- Enforced role-based access (Police, Court, Admin) in the smart contract using Solidity modifiers, covered by contract and encryption test suites

**COPREPER — AI-Powered Interview Prep Platform** | *Next.js, Node.js, Express, SQLite, JWT, OpenAI/Claude* Live 2025

- Built a full-stack platform for CS students to document projects and practise technical interviews with AI-evaluated answers
- Implemented JWT auth with secure httpOnly cookie sessions and SQLite FTS5 full-text search in a three-tier architecture
- Designed an Adapter Pattern so AI providers (OpenAI/Claude) can be swapped without changing application code

## Achievements & Leadership

- **1st Place, Hackathon — TechFest 2026** (BTKIT): built and presented Nishaan solo after teammates were absent
- **2nd Place**, inter-college Aerobotics competition and **3rd Place**, Project Exhibition — TechFest 2026
- **Innovation Cell Lead** (2024 – Present): organized the college-level Smart India Hackathon 2025 end to end and coordinated the annual coding fest and technical events

## Technical Skills

**Languages:** TypeScript, JavaScript, Python, C++, SQL, Solidity
**Web:** Next.js, React, Node.js, Express.js, Tailwind CSS, REST APIs
**Cloud & AI:** AWS (Lambda, Step Functions, DynamoDB), LLM APIs (OpenAI, Claude), PyTorch, Transformers (BERT)
**Tools & Databases:** Git/GitHub, Docker, PostgreSQL, MongoDB, SQLite
**Certifications:** MongoDB University (AI Agents, RAG, Schema Design); Postman API Fundamentals Student Expert